SCORE: 2022

HIGH-SCORE: 2022

LET'S PLAY CHESSBOT

MENU

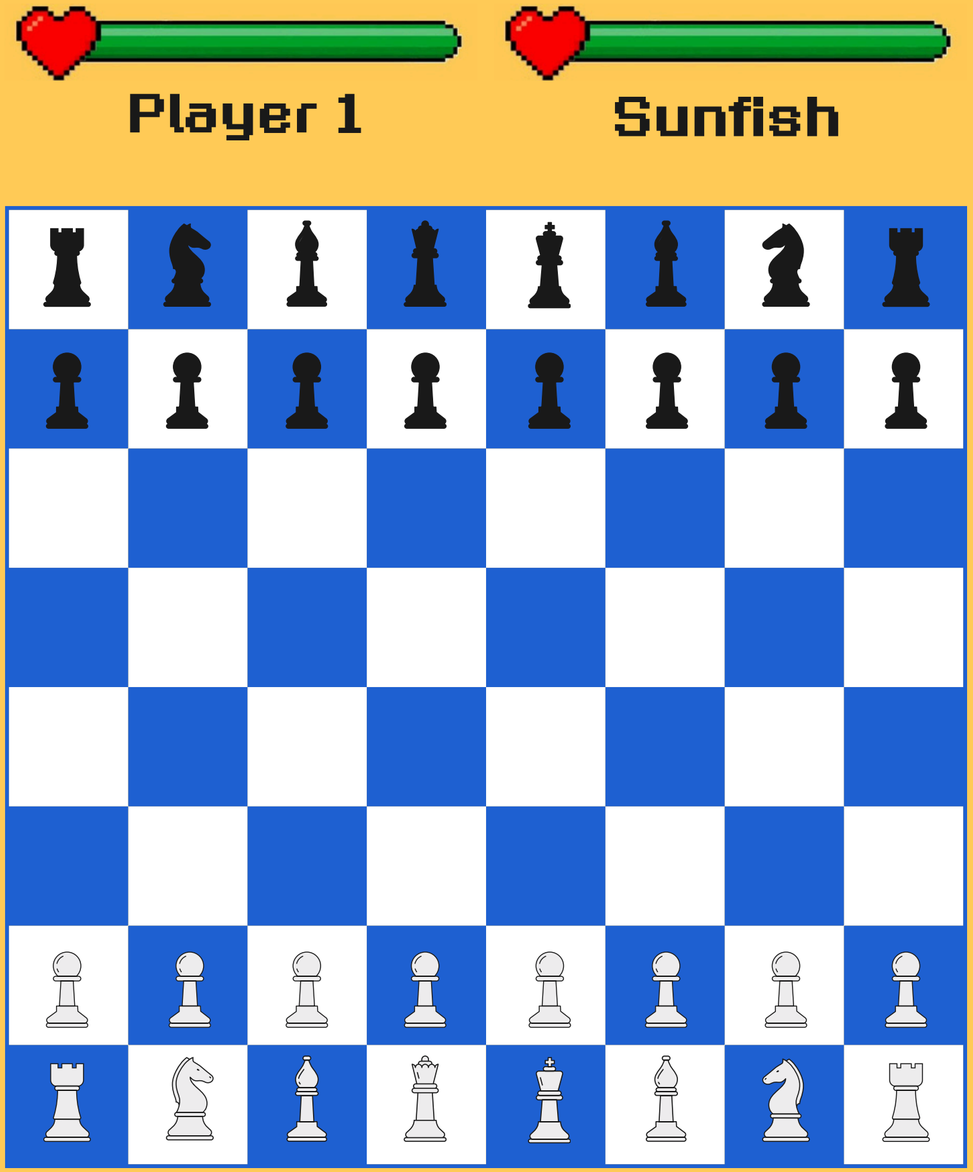
SIGN IN





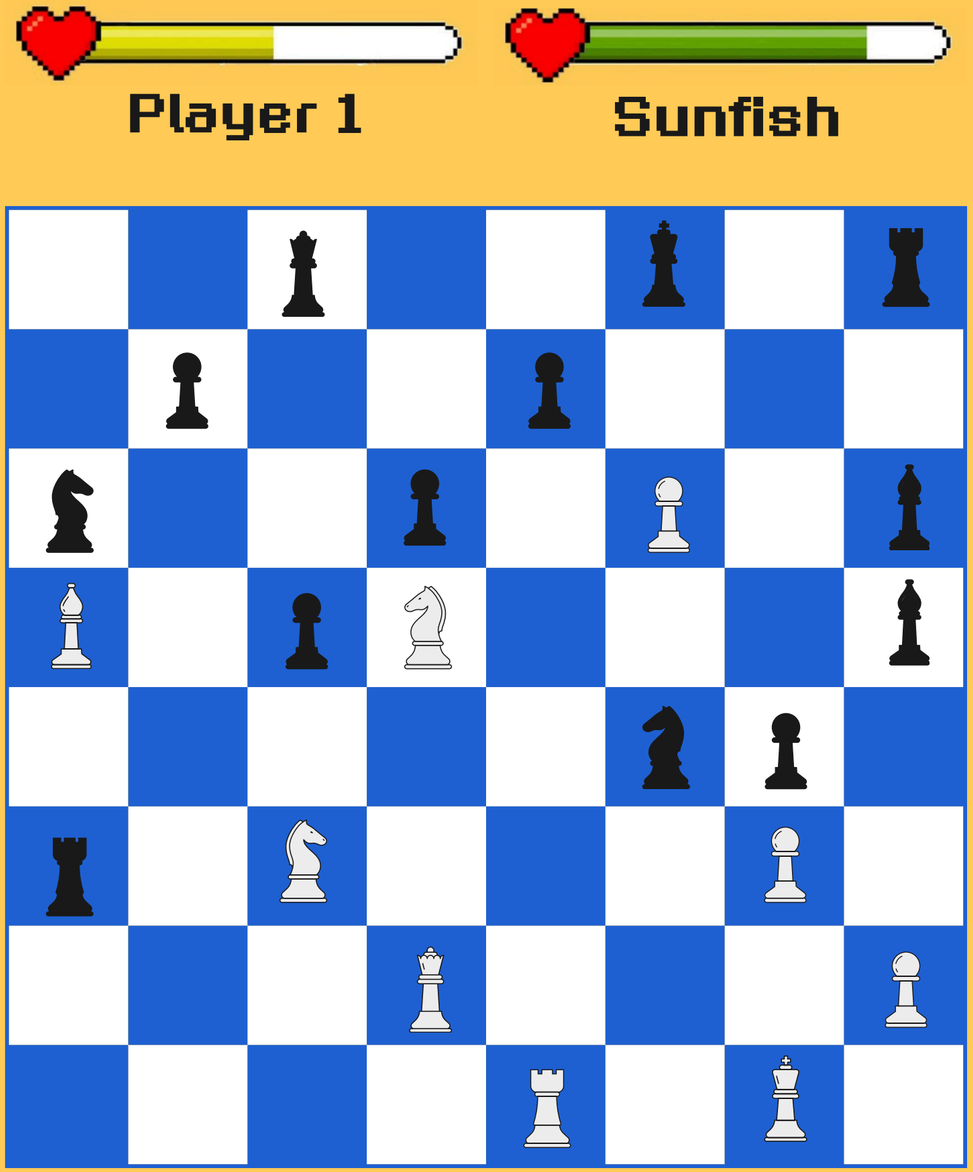
CHESSBOT'S CONCEPT:

The idea behind ChessBot is simple: to create an automated system capable of playing chess against a human opponent. What makes this concept innovative, and the project particularly interesting, is ChessBot's ability to interact with a physical chessboard, picking up, moving and removing pieces directly from it.



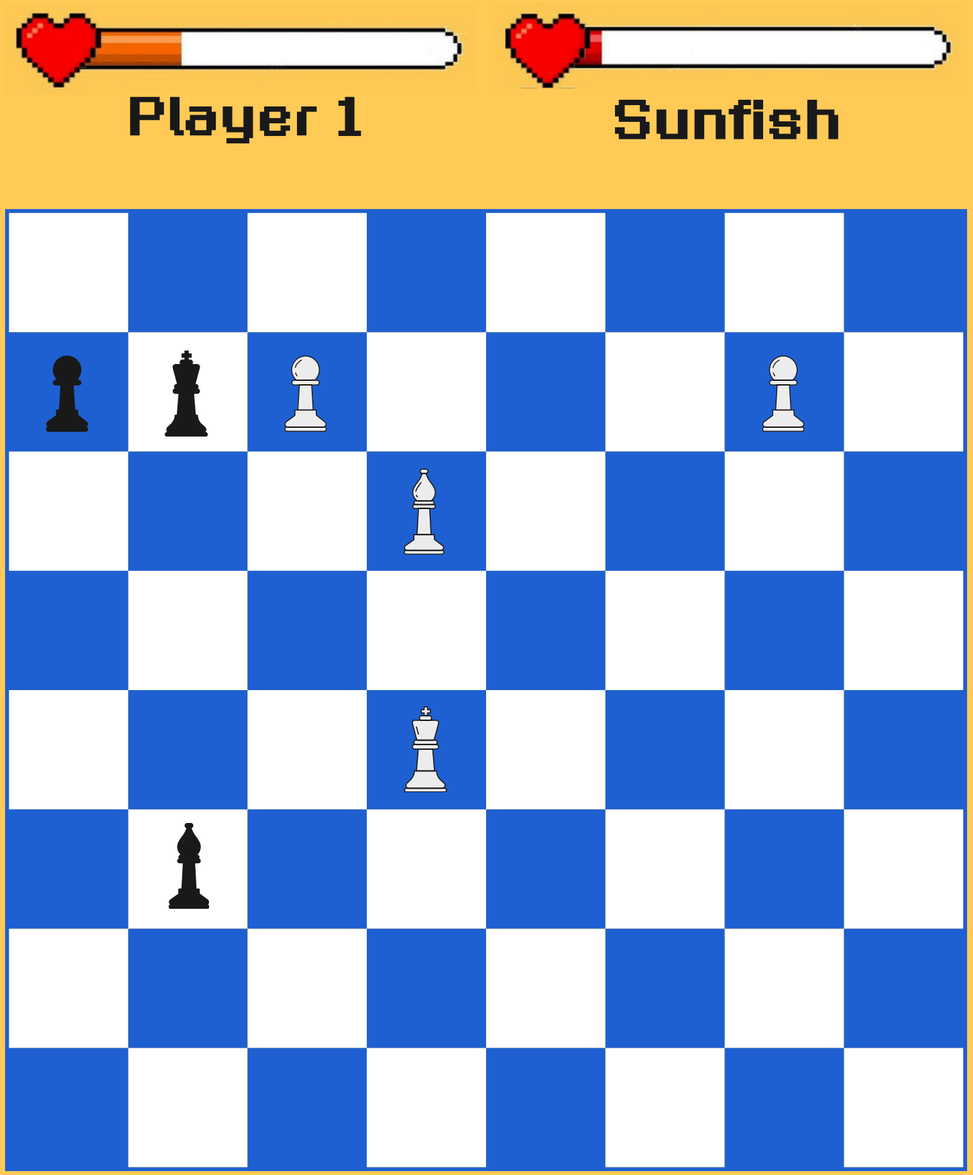
CHESSBOT'S DESIGN:

The brain behind our project is the Sunfish chess engine, which coordinates and oversees the entire game. Players can conveniently select their moves through a clear and intuitive user interface that recreates the mechanics of chess. Before making its own move, Sunfish validates the opponent's move and then signals the PLC to execute the inverse kinematics of the three axes.



CHESSBOT'S REALIZATION:

The first steps in developing ChessBot involved assembling and wiring the components of the three-axis system. The electrical cabinet, which had already been partially wired, was completely dismantled and rebuilt according to more efficient design principles.
After testing the motor drivers, we developed the PLC program responsible for controlling the physical chessboard, moving the pieces and capturing the opponent's pieces.
The final step was implementing OPC UA communication with the chess engine, developed in Python, enabling information exchange between the different systems.



THANK YOU FOR PLAYING!

PLAY AGAIN?

YES

NO

